

Raj Modi

Python & AI Developer | LLMs & Multi-Agent Systems · Engineering-Process Automation · ML · FastAPI

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SUMMARY

Final-semester B.Tech (CSE — AI & Data Science) engineer who builds **Python tools and LLM-powered systems that automate manual engineering and SDLC workflows** — **LLM code generation** (natural-language→SQL), **automated documentation generation, test automation** (pytest / Vitest / Selenium, CI-gated), **multi-agent LLM orchestration** (LangGraph), and **RAG / MCP**. Delivered a Python automation engine for a **real automotive client (VW Group Academy)** that cut manual effort ~80% — a direct fit for using **Python + LLMs to turn manual SDLC effort into measurable cost savings**.

TECHNICAL SKILLS

Languages: Python, TypeScript / JavaScript (ES6+), SQL, Java
AI / LLM: LLMs (LLaMA 3.3 70B / Groq, Gemini, Claude), LangGraph multi-agent orchestration, Hybrid RAG (dense + BM25 + RRF), HyDE, CRAG, MCP, Pydantic structured output, prompt engineering, spaCy NLP
ML & Automation: scikit-learn (Gradient Boosting, Isolation Forest), SHAP, PSI drift, 5-fold CV, Holt-Winters, Prophet / PyTorch, pandas / NumPy / SciPy; Python ETL (openpyxl), constraint-satisfaction heuristics, Tkinter, PyInstaller
Back-End, Test & DevOps: FastAPI, Flask, REST, WebSockets, SSE, tRPC, Node.js; pytest, Vitest, Selenium; Docker / Compose, GitHub Actions CI/CD; PostgreSQL, DuckDB, pgvector, Redis

PROJECTS

Škoda VWS Room Booking – Constraint-Based Training-Schedule Automation Engine Jun 2025 – Feb 2026
8-Month Industrial Project | **VW Group Academy (Škoda / VW)** | **Team Lead** | Python, pandas, NumPy, openpyxl, Tkinter, PyInstaller, Constraint-Satisfaction / Greedy Heuristics,

ETL

- Architected and shipped a **production desktop app** for a **real automotive client** — the **Volkswagen Group Academy (Škoda / VW / Audi / Porsche)** — **leading a 3-member team** over an 8-month engagement; **fully automates annual training-calendar allocation** of ~1,243 events across 4 centres (Pune, NCR, Bengaluru, Hyderabad) previously scheduled by hand.
- Built a **pandas / openpyxl ETL pipeline** normalising heterogeneous *Training Calendar / Room Booking* workbooks, then a **constraint-satisfaction greedy first-fit allocator** with **preference-tiered room assignment** (Classroom → Cluster → Workshop), **shift-level temporal partitioning**, and **hard no-double-booking conflict detection** — hitting **98.5% auto-allocation (1,224 / 1,243)** and **~80% less manual effort**, with an auto-generated **capacity-utilization matrix**; shipped as a **single-click PyInstaller executable** with a Tkinter GUI.

Verita – AI-Native Financial-Crime & Compliance Intelligence Platform 2026
Real-ML Risk Scoring | **Agentic Auditable Investigator** | **NL→SQL** | Python, FastAPI, scikit-learn, DuckDB, SciPy, SHAP, joblib, pytest, GitHub Actions CI, React, TypeScript,

ECharts

- Built a full-stack **FastAPI + React/TypeScript** platform whose **FCC Risk Engine** trains a **scikit-learn ensemble (Gradient Boosting + Isolation Forest)** on the **real ULB credit-card fraud corpus** (284K txns, 0.17% positive class) — **honest held-out ROC-AUC 0.913 / PR-AUC 0.65**, 5-fold CV, SHAP + permutation importance, **PSI drift monitoring**, a live **decision-threshold control**, and a ranked **AML alert queue**.
- Engineered an **autonomous “Auditable Investigator” agent** that plans AML hypotheses, validates each against a live **DuckDB** query, and **hash-chains the reasoning trace with SHA-256** (each step folding in the prior hash) for a **tamper-evident, reproducible record** — plus a **natural-language→SQL console**, **Holt-Winters forecasting** (backtested MAPE), and an **82-test suite (72 pytest / 10 Vitest)** gated by GitHub Actions CI.

AgentForge – Production Multi-Agent LLM Orchestration Engine 2026
7 Autonomous Agents | **LangGraph State Machine** | **Automated Documentation** | Python 3.11, FastAPI, LangGraph, Claude / Gemini, Pydantic, SQLAlchemy, PostgreSQL,

Redis, WebSockets, Docker Compose, MCP, JWT

- Orchestrated **7 autonomous LLM agents** via a **LangGraph state machine** in a **9-node “diamond” topology** with **parallel fan-out/fan-in** (4 concurrent agents) and a multi-round **“Board Meeting” debate** for adversarial cross-validation; each agent runs a **Generate → Self-Critique → Refine** chain-of-thought with **Pydantic-validated, schema-safe JSON**, automating **30-page report / documentation generation**.
- Streamed real-time agent reasoning over **WebSockets**; secured with **JWT + RBAC**, persisted via **SQLAlchemy / PostgreSQL + Redis**, exposed agent tools through a custom **MCP server**, and packaged the stack for **one-command Docker Compose** deployment.

Nyay AI – Multilingual Legal-Intelligence Platform with Hybrid RAG 2026
Hybrid Retrieval | **LLaMA 3.3 70B** | **8-Agent Studio** | **22+ Languages** | Python, Flask, LLaMA 3.3 70B (Groq), sentence-transformers, BM25, NER, RRF, HyDE, CRAG, D3.js,

SSE

- Engineered a **hybrid RAG pipeline** fusing **dense embeddings (all-MiniLM-L6-v2)**, a **custom legal BM25 scorer**, and **NER entity-overlap** via **Reciprocal Rank Fusion** — with **HyDE query expansion** and **CRAG self-correction** triggering a **confidence-gated web-search fallback** before **LLaMA 3.3 70B (Groq LPU)** generation, streaming **source-cited answers** over **SSE** in **22+ Indian languages**.
- Built an **8-agent Studio** (adversarial AI Courtroom, clause-redlining Adversarial Drafter, Risk Heatmap, Legal-Twin persona), a **D3.js force-directed knowledge graph** (Acts / Sections / Courts / Cases), and a real-time **RAG Pipeline Observatory**.

EDUCATION

B.Tech, Computer Science & Engineering (AI & Data Science) 2022 – 2026
MIT World Peace University (MIT-WPU), Pune

ACHIEVEMENTS & CERTIFICATIONS

- **SIH 2025 – Rank 63 (Top 10%)**, Smart India Hackathon | **AIR 8**, National Creative Olympiad | **Google Advanced Data Analytics & IBM Data Visualization (Python) certified** | **Joint President**, SARS MIT-WPU; **Execution Head**, RIDE 2025 (500+ attendees).